

Yeongbeom **Song**

Solution Architect · Software Architect

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"Always be creative, always keep growing."

Education

Ajou University

Suwon, S.Korea

B.S. in Software and Computer Engineering — *Mar. 2020 - Feb. 2027(Expected)*

- Total GPA: 4.44 / 4.50, Major GPA: 4.50 / 4.50(Completed 6 semesters, Credits earned: 122 / 140).
- National Scholarship for Science and Engineering – awarded to promising students in STEM fields.

Sunrin Internet High School

Seoul, S.Korea

Diploma in Department of Software — *Mar. 2017 - Feb. 2020*

- National Competency Standards(NCS) Straight-A in Software Department.
- Sunrin Academy Scholarship - awarded to promising students in Software Department.

Research Interests

Ubiquitous and Distributed Systems	Developing real-time services across mobile, wearable, and distributed IoT systems
Human-Centered Embedded & IoT Systems	Designing embedded & IoT systems with a focus on user experience and usability
Resource-Constrained Embedded Optimization	Optimizing performance, power & latency under limited hardware resources

Work Experience

Defense Integrated Data Center, Ministry of National Defense

Gyeryong, S.Korea

Security & System Operator(Military Service) — *Jul. 2024 - Feb. 2026(Expected)*

- Served mandatory military service(Air Force Cyber Systems Specialist) as Security & System Operator at the DIDC, MND.
- Operated and maintained military computer systems, including deployment, fault detection, and error analysis on Linux platforms.
- Conducted cyber defense operations, including threat detection, incident response, and security incident analysis.
- Performed rack inspections and network equipment examination to ensure the stability of large-scale systems.

Vestella Lab Inc.

Anyang, S.Korea

Intern Researcher — *Dec. 2023 - May. 2024*

- Conducted advanced research and technology enhancement for indoor autonomous driving as an intern researcher at Vestella Lab.
- Improved beacon scanning and positioning in Linux-based embedded systems for real-time indoor localization.
- Enhanced real-time capability, throughput, and embedded system compatibility of beacon scanning modules.
- Increased indoor localization accuracy and precision through optimization of beacon placement and parameters.

EY Labs(Startup)

Seoul, S.Korea

Founder & Software Architect — *Apr. 2022 - May. 2024*

- Founded a startup with a ₩50M seed grant from the SW Maestro Sustainable Growth Program, MSIT.
- Led the design, development, and operation of "Easy Yum: Personalized Diet Plan Recommendation and Management Service".
- Developed the client application using the Flutter framework and the server application using the Spring Boot framework.
- Built a high-availability, fault-tolerant infrastructure on AWS and Firebase, implementing CI/CD for efficient operations.

AllCareDX(Startup)

Seoul, S.Korea

Software Engineer — *Feb. 2023 - Mar. 2023*

- Developed and deployed 'Highsleep', a sleep management application and healthcare device, as a software engineer at AllCareDX.
- Improved and optimized the software's architecture, enhancing performance and stability, and performed preprocessing for deployment.
- Implemented and tested Bluetooth Low Energy(BLE) communication and integration between the application and healthcare device.

Sunrin Internet High School

Seoul, S.Korea

Adjunct Instructor — *May. 2022 - Aug. 2022*

- Served in advanced major Courses at Sunrin Internet H.S., teaching application development and software engineering methodologies.
- Delivered lectures covering the cross-platform application design, development, and deployment using the Flutter framework.
- Taught Agile methodologies, including XP, Scrum, and Lean, with hands-on practice and project-based instruction.

Project Experience

Major Projects

Indoor Beacon Scanning & Localization Performance Improvement on Embedded Linux

Professional Project

Embedded Engineer — *Dec. 2023 - May. 2024*

- Developed a low-level C-based beacon scanning module utilizing the native Linux Bluetooth protocol stack(BlueZ).
- Designed a multithreaded architecture to parallelize scanning, filtering, and transmission, employing mutexes to handle concurrency issues.
- Implemented a named-pipe interface for beacon data exchange across modules and a positioning module leveraging scanned beacon data.
- Reduced latency by over 90% compared to in-house modules, establishing a stable environment for internal indoor autonomous driving tests.

Easy Yum : Personalized Diet Plan Recommendation and Management Service

Professional Project

Software Architect — *Apr. 2021 - Apr. 2024*

- Launched and operated Easy Yum, EY Labs' startup product, on Google Play Store and Apple App Store for three years.
- On the frontend, developed a custom design system, tutorial, version check, forced update, advertising, and payment modules.
- On the backend, developed a cloud infrastructure, nutrition database, diet recommendation algorithm, and authentication modules.
- Achieved peak MAU 4,200, WAU 1,800, DAU 500, and secured ₩50M in preliminary startup funding during the operation period.

Alledin : Restaurant Recommendation Service for Food Allergy Management

Academic Project

Software Architect — *Mar. 2023 - Dec. 2023*

- Developed Alledin in an undergraduate course, released and maintained on Google Play and Apple App Store.
- Served as project leader, handling planning, design, and build with a focus on high availability and fault-tolerant infrastructure.
- Built Korea's first database integrating fragmented franchise brand allergy data by menu, enabling comprehensive allergy management.

HighSleep: A real-time sleep-stage analysis–based sleep-inducing healthcare system

Professional Project

Software Engineer — *Mar. 2023 - Dec. 2023*

- Implemented BLE-based device pairing and control for the healthcare device(ESP32), enabling reliable mobile–device interaction.
- Improved core logic and resolved usability issues to refine interaction flow and deliver a smoother, more consistent user experience.
- Developed device discovery, connection, and actuator control features, integrating real-time sensor data through custom BLE modules.
- Enhanced system stability through refactoring, dependency updates, and focused debugging across embedded and mobile components.

Smart Drainage & Manhole Devices for Prevention of Heavy Rainfall and Flooding

Academic Project

Embedded Developer — *Sep. 2022 - May. 2023*

- Developed in an undergraduate course, the system used a client-server architecture with multiple Raspberry Pis connected to various sensors.
- Implemented sensing, processing, communication, and control in a multi-threaded environment via a custom socket-based protocol.
- Managed hardware setup, software development, collaboration templates, and system design as project leader.

Intelligent Mask Sterilization Device and Mask Management Service

Competition Project

Embedded Developer — *May. 2020 - Sep. 2020*

- Developed during a university capstone competition, integrating multiple sensors, a NodeMCU-based device, and a mobile application.
- Implemented optimized software for low-power, low-performance environments, enabling seamless integration with the mobile application.
- Designed hardware using perfboard and 3D printing to complete a prototype, and conducted performance and stability testing.

Other Projects

2025	**Startup Stove League : Circular Entrepreneurship Analysis and Matching Platform**, Ideated	*Competition Project*
2025	**Frenzip : AI-Agent Based Rental Property Management Automation Platform**, Ideated	*Competition Project*
2024	**Ultra-Localized & Short-Term Heavy Rainfall and Flooding Monitoring System**, Ideated	*Competition Project*
2023	**LLM-based Intelligent Patent Search Engine for Analysis and Utilization**, Ideated	*Competition Project*
2023	**Cross-Platform Eye Usage Analysis and Eye Health Management Service**, Ideated	*Competition Project*
2022	**Social Media Trend-Based Photo Spot and Pose Recommendation Service**, Ideated	*Competition Project*
2021	**KTEEN : Welfare Curation Services for Out-of-school Youth**, Ideated and Developed	*Competition Project*
2021	**Ajou Swift : Chrome Extension for Ajou University Students**, Ideated and Developed	*Personal Project*
2020	**Video Conferencing Support Platform for Active Communication**, Ideated	*Competition Project*
2019	**Smart Band Device for Detecting and Preventing Heat-Related Illnesses**, Ideated and Developed	*Competition Project*

Publications & Patents

Publications

Case study on usability increase through UI/UX continuity in Flutter Framework

KICS Winter Conference

Yeongbeom Song, Hwanyong Lee — *Jan. 2024*

- Flutter is a cross-platform framework that provides the same UI across multiple platforms with one code base to maintain UI/UX continuity.
- Flutter-based applications show that they can maintain UI/UX continuity, improving both learnability and usability.

Patents

2024	**10-2024-0121524**, Personalized Food Information Providing System and Method(Pending)	*KR Patent*
2024	**10-2024-0026170**, Eye Health Solution Device for Modern Lifestyles(Pending)	*KR Patent*
2024	**10-2024-0025659**, Smart Flood Prevention System(Pending)	*KR Patent*
2023	**10-2023-0002124**, Photo Spot Recommendation Using SNS Data(Pending)	*KR Patent*
2021	**10-2021-0141568**, Diet Recommendation System(Rejected)	*KR Patent*

Awards & Honors

Awards

2024.10	**Excellence Prize(2nd place)**, 6th Air Force Hackathon Competition	*ROKAF, KoEF*
2024.06	**Honorable Prize(4th place)**, 2nd LINC 3.0 Startup Idea Competition	*Ajou University*
2024.06	**Outstanding Prize(3rd place)**, 2024-1 BM/Invention Design Competition	*Ajou University*
2023.12	**Special(Chief of Staff) Prize**, 19th TOPCIT Award for Top Performers	*ROKAF, IITP*
2023.12	**Excellence Prize(2nd place)**, 2023-2 Invention Idea Competition	*Ajou University*
2023.12	**Pilot Prize(5th place)**, 2023-2 Blue Semester Program Competition	*Ajou University*
2023.05	**Excellence Prize(2nd place)**, 2023-1 Invention Idea Competition	*Ajou University*
2023.05	**Outstanding Prize(3rd place)**, 2023-1 Invention Idea Competition	*Ajou University*
2022.12	**Excellence Prize(2nd place)**, 2022-2 Invention Idea Competition	*Ajou University*
2022.08	**Honorable Prize(4th place)**, 4th Youth Ideathon Competition	*Ajou University*
2021.12	**Honorable Prize(4th place)**, 7th LINC+ Startup Idea Competition	*Ajou University*
2021.09	**Grand Prize(1st place)**, 2021 Technology Sharing Capstone Design Competition	*Ajou University*
2021.09	**Outstanding Prize(3rd place)**, 2021 Convergence Capstone Design Competition	*Ajou University*
2020.12	**Excellence Prize(1st place)**, 5th LINC+ Startup Idea Competition	*Ajou University*
2020.10	**Excellence Prize(1st place)**, 6th Uiwang Youth Policy Proposal Competition	*Uiwang City*
2020.09	**Outstanding Prize(2nd place)**, 2020 Technology Sharing Capstone Design Competition	*Ajou University*
2019.02	**Excellence Prize(2nd place)**, 1st Maker Heroes in Korea Student Competition	*Code Club Korea*
2018.10	**Honorable Prize(3rd place)**, Seoul Maker 2018 Exhibition & Competition	*Seoul Digital Fnd.*
2018.03	**Honorable Prize(3rd place)**, 1st Design Thinking Hackathon Competition	*Net Impact Korea*
2017.12	**Outstanding Prize(2nd place)**, 14th AppJam Hackathon Competition	*Asan Nanum Fdn.*

Honors

2023.12	**Dean's List(2023-2)**, Recognized for outstanding academic performance during the semester	*Ajou University*
2023.01	**Certificate of Commendation**, Recognized for contributions to mentoring culture and future education	*Yongsan District*
2023.01	**Certificate of Recognition**, Recognized for excellence in self-directed learning and development	*Ajou University*
2022.12	**Dean's List(2022-2)**, Recognized for outstanding academic performance during the semester	*Ajou University*
2022.04	**Certificate of Commendation**, Recognized for contributions to the development of youth policy in Suwon	*Suwon City*
2021.10	**Certificate of Recognition**, Recognized for excellence in self-directed learning and development	*Ajou University*
2021.06	**Dean's List(2021-1)**, Recognized for outstanding academic performance during the semester	*Ajou University*
2020.12	**Dean's List(2020-2)**, Recognized for outstanding academic performance during the semester	*Ajou University*

Scholarships

2023.08	**National Scholarship for Science and Engineering**, Awarded to top STEM undergraduates nationwide	*MSIT, KOSAF*
2021.08	**Guwon Scholarship(2021-2)**, Awarded to outstanding SCE undergraduates at Ajou University	*Guwon Fnd.*
2020.08	**Yulgok Scholarship(2021-2)**, Awarded to top performers within the SCE department at Ajou University	*Ajou University*
2021.02	**Guwon Scholarship(2021-2)**, Awarded to outstanding SCE undergraduates at Ajou University	*Guwon Fnd.*
2021.02	**Daewoo Scholarship(2021-1)**, Awarded to top performer within the software college at Ajou University	*Ajou University*
2020.08	**Yulgok Scholarship(2020-2)**, Awarded to top performers within the SCE department at Ajou University	*Ajou University*
2019.05	**Sunrin Academy Scholarship**, Awarded to outstanding performers at Sunrin Internet High School	*Sunrin Internet H.S.*

Skills

Programming	C/C++, Python, Java, Dart, JavaScript, SQL
Embedded & IoT	Embedded Linux, Raspberry Pi, NodeMCU, Arduino, BLE
Systems & Cloud	Linux, AWS, Firebase, Docker, CI/CD
Front-end	Flutter, React, Web Development(HTML/CSS)
Back-end	Spring Boot, Express, Django, MySQL, REST

Certifications

English proficiency

2025.06 **TOEIC 720**, Educational Testing Service(ETS) *Valid*

Certifications

2025.07 **Linux Master Level 2**, Korea Association for ICT Promotion(KAIT) *Valid*
2025.06 **Engineer Information Processing**, Human Resources Development Service of Korea(HRDK) *Valid*
2025.05 **TOPCIT Level 4(810)**, Institute of Information & communications Technology Planning & Evaluation(IITP) *Valid*
2024.12 **SQL Developer**, Korea Data Agency(K-DATA) *Valid*
2023.07 **COS PRO Level 1(C)**, YBM IT TEST *Valid*
2021.12 **Solutions Architect - Associate(SAA-C02)**, Amazon Web Services(AWS) Certified *Expired*
2019.07 **Craftsman Information Processing**, Human Resources Development Service of Korea(HRDK) *Valid*

Extracurricular Activity

Software Maestro Expert by Ministry of Science and ICT *Seoul, S.Korea*

Tutor(13, 14th) *Jun. 2022 - Nov. 2023*

- Selected as a tutor in the SW Maestro program by the MSIT, providing guidance and mentoring to trainees.
- Shared insights on the training process and provided feedback on computer science and engineering studies.
- Served as Scrum Master for the team, facilitating the effective adoption and application of Agile methodologies.

University Student Mentoring Project in Yongsan *Seoul, S.Korea*

Mentor *Sep. 2022 - Dec. 2022*

- Provided 1:1 advanced major mentoring to high school students in software studies.
- Conducted code reviews, refactoring, and pair programming focused on mobile application development.

Software Maestro Program by Ministry of Science and ICT *Seoul, S.Korea*

Trainee(12th) *Feb. 2021 - Jan. 2022*

- Selected as a trainee in the SW Maestro program by the MSIT, completing the full curriculum.
- Enhanced software expertise through apprenticeship-based training with researchers in computer science and engineering.
- Completed the full software development lifecycle, gaining software engineering skills demanded by the industry.
- Developed and launched 'Easy Yum', while undertaking preparations to establish 'EY Labs' during the program.

Chorok, Community Problem Solving Start-up Club in Ajou univ. *Suwon, S.Korea*

Member *Mar. 2021 - Dec. 2021*

- Participated as a member of a startup club addressing local community issues and conducting social contribution activities.
- Organized a youth ideathon to address community challenges and submitted selected proposals to the City of Suwon.
- Delivered software classes using micro:bit to elementary school students in Suwon for community outreach.

kkum-iji, A college student mentor group in Gunpo *Gunpo, S.Korea*

Mentor(6, 7th) *Mar. 2020 - Dec. 2021*

- Delivered lectures introducing academic majors and career paths to middle school students.
- Conducted group mentoring sessions on career exploration and occupational choices for middle school students.